

This workbook has both a paper and a Jupyter notebook submission.

1. Exercise 5.17
2. Using automatic differentiation, create a program to calculate the first and second derivatives for the function:

$$f(x) = x^2 \cos\left(\frac{2\pi x}{L}\right) - 0.5x \quad (1)$$

Derive your state vectors with the necessary operation in your paper submission. Test your code for values $x = 2, 5, 8$ for $L = 2$.